

Rotaviruses

**Background document for the
WHO Guidelines for drinking-water quality and the
WHO Guidelines on sanitation and health**

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Highlights

- There is strong evidence linking transmission of rotaviruses with unsafe drinking-water, inadequate sanitation and poor hygiene. Rotaviruses are major waterborne pathogens. Several waterborne outbreaks have been documented from inadequately disinfected drinking-water supplies.
- Humans are the primary reservoir for strains of rotavirus that cause human disease. There is some evidence that animal rotavirus strains can infect humans, suggesting a potential for zoonotic transmission.
- Transmission is primarily by the faecal–oral route. Rotaviruses are spread through person-to-person contact and through contaminated food, water and fomites.
- Infections may be asymptomatic (especially in older children and adults) or cause acute disease characterized by an abrupt onset of severe watery diarrhoea with vomiting, fever and/or abdominal pain. The outcome may be fatal if the infection is not appropriately treated, especially in infants and underweight children.
- The burden of diarrhoeal disease caused by rotavirus infection among infants and young children is extremely high; it infects nearly every child worldwide by 5 years of age. The rotavirus disease burden is higher in low-income than in high-income settings. The vast majority of rotavirus deaths occur in low-income settings. In temperate regions, illnesses peak in winter and spring months. Rotavirus reinfections are quite common throughout life, although with each reinfection, the illness tends to become milder.
- Rotaviruses are highly infective with an estimated ID₅₀ (50% infective dose) of 6 focus-forming units.
- No standard cell culture method is available for testing environmental samples, but molecular-based detection methods are available.
- Rotaviruses are commonly present in raw sewage, treated wastewater and surface waters and have been detected in groundwater and drinking-water. They can persist for long periods of time in water.
- Systematic approaches that identify and control faecal contamination along the whole sanitation service and drinking-water supply chains are needed to manage microbial risks, including from rotaviruses. However, as they spread easily from person-to-person, other control measures, including adequate hygiene, are required.
- Reduction of rotaviruses by wastewater and drinking-water treatment varies depending on the process. Although rotaviruses are moderately resistant to disinfection with chlorine, it can be effective provided adequate doses and contact times are applied.
- Due to the smaller size of rotaviruses relative to *E. coli*, resulting in lower removal by treatment processes such as filtration, and their greater environmental persistence, the absence of *E. coli* (or alternatively, thermotolerant coliforms) is not a suitable indicator of the absence of human rotaviruses in drinking-water. However, the presence of *E. coli* is a suitable indicator of the possible presence of human rotaviruses.

1 General description

Rotaviruses belong to the genus *Rotavirus*, within the family Reoviridae. They consist of a segmented double-stranded RNA genome in a non-enveloped icosahedral capsid with a diameter of 50 nm to 65 nm. This capsid is surrounded by a double-layered shell with the appearance of a wheel, hence the name rotavirus. The diameter of the entire virus is about 100 nm (Estes & Greenberg, 2013; da Silva et al., 2016).

There are currently nine known species of rotaviruses designated as A through D and F through J, each of which consists of a number of subgroups (ICTV, 2021). Species A, B, C and H can infect humans, with species A being the most commonly diagnosed cause of rotavirus infections in both low- and high-income countries (Estes & Greenberg, 2013; da Silva et al., 2016). Rotavirus A is further divided into serotypes using characteristics of the outer capsid proteins VP7 and VP4. Several unique types of VP7 (G types) and VP4 (P types) are currently known, demonstrating great antigenic diversity (Mijatovic-Rustempasic et al., 2016; Suzuki, 2019).

2 Human health effects

Rotavirus infection is most common in infants and children under 5 years of age (Parashar et al., 2006; Arakaki et al., 2021). In children, an acute rotavirus infection is characterized by an abrupt onset of severe watery diarrhoea with vomiting, fever and/or abdominal pain (Estes & Greenberg, 2013; Heymann, 2015). Dehydration and metabolic acidosis may develop due to loss of fluids and electrolytes; the outcome may be fatal without appropriate treatment, especially in infants and underweight children (Wickelgren, 2000). Fluid replacement is critical in the treatment of severe rotavirus infections. Older children and adults can also be infected by rotaviruses; adults tend to have similar but milder symptoms compared to children. Asymptomatic infections with rotaviruses can occur in older children and adults (Anderson & Weber, 2004). Rotavirus reinfections are quite common throughout life (even in high income countries), although with each reinfection, the illness tends to become milder (Crawford et al., 2017).

3 Epidemiology

The burden of diarrhoeal disease caused by rotavirus infection among infants and young children is extremely high. It infects nearly every child worldwide by 5 years of age (Estes & Greenberg, 2013; da Silva et al., 2016). For children in low-income settings, rotavirus is the leading cause of moderate-to-severe diarrhoea in the first year of life (Kotloff et al., 2013; Liu et al., 2016). In higher income settings with temperate climates, rotavirus infection shows a distinct seasonality with peaks in the winter and spring months (Patel et al., 2013). In low-income, typically tropical settings, there is usually a background of year-round circulation punctuated by episodes of intense transmission (Patel et al., 2013) but recent studies suggest that some seasonality may occur with peaks during cooler and drier months (Chao et al., 2019). Differences in transmission patterns, health care access, childhood co-infections and co-morbidities, drive substantial differences in the rotavirus disease burden between low- and high-income settings (Levy et al., 2009; Patel et al., 2013).

Despite the availability of prequalified rotavirus vaccines since 2008 (WHO, 2021), rotavirus remained the leading cause of diarrhoeal disease mortality worldwide in 2019, accounting for almost 30% of all diarrhoea deaths among children under five years of age (GBD, 2020). The vast majority of rotavirus deaths occur in low-income settings (Tate et al., 2016, GBD 2020). However, the relative importance of rotavirus in contributing to the burden of infant diarrhoeal disease is expected to decrease due to the use of the rotavirus vaccines (Platts-Mills et al., 2015, WHO 2021).

4 Transmission

Human rotaviruses are transmitted primarily by the faecal–oral route and can be spread by direct person-to-person contact and through contaminated food, water and fomites (Estes & Greenberg, 2013; Heymann, 2015; da Silva et al., 2016). Person-to-person contact appears to be the most important source of transmission. Airborne spread has been hypothesized but investigations into the viability of aerosolized rotaviruses are needed to confirm airborne transmission (Omatola et al., 2022).

Humans are the main reservoir for rotavirus strains that cause disease in people. A diverse range of wild and domestic animals can be infected with animal rotavirus strains, and there is increasing evidence that some animal rotaviruses can infect humans, either directly or through gene exchange and reassortment with human rotaviruses (Muller & Johne, 2007; da Silva et al., 2016), suggesting they are a candidate for zoonotic transmission. However, human strains remain the major risk at this time. The interaction between rotavirus and the histo-blood group antigens (HBGAs) is strain-specific and has been observed in vitro as well as in population-based studies (Nordgren et al., 2014; Van Trang et al., 2014). Differences of HBGA profiles in the population may contribute to differences in susceptibility, rotavirus disease burden, genetic diversity of circulating strains and vaccine effectiveness.

Rotaviruses are excreted in large quantities by infected individuals (4.5×10^7 to 3.2×10^{12} genome copies (GC)/g of faeces) from 2 days before and up to 10 days after symptomatic illness (Estes & Greenberg, 2013; Heymann, 2015). The relative infectivity of rotavirus is high with an estimated ID₅₀ (50% infective dose, i.e. the number of organisms required to infect 50% of exposed subjects) of 6 focus-forming units (Ward et al., 1986).¹ There is also evidence of a strong quantity-dependent association with diarrhoeal symptoms (Liu et al., 2016). The incubation period can range from 24 to 48 hours (Estes & Greenberg, 2013). Vaccination is effective in reducing the risk of infection but is not 100% protective (Burnett et al., 2020). Natural infection also does not provide full protection from future infections (Fischer et al., 2002).

¹ From experiments with human volunteers, from epidemiological evidence and from experimental animal studies. Based on the criteria established by WHO in the Guidelines for drinking-water quality as follows: High means infective doses can be 1–10² organisms or particles, Moderate 10²–10⁴ and Low > 10⁴.

5 Detection methods

Wild-type strains of rotavirus A are not readily grown in cell culture. There is no standard culture method for the detection of rotaviruses in environmental samples, but a number of molecular-based methods are available (Haramoto et al., 2018; Corpuz et al., 2020). Distinguishing between viable and non-viable viruses is a shortcoming of molecular methods. Combining cell culture infectivity and molecular methods makes it possible to improve the detection of infectious rotaviruses (Rutjes et al., 2009; Li et al., 2010). Due to their low levels in the water environment, concentration methods such as ultrafiltration are needed to detect rotaviruses in water (Grassi et al., 2010; Corpuz et al., 2020).

6 Source and occurrence

Various environmental matrices have been examined for rotaviruses. A summary of their sources, concentrations and persistence is provided in Table 1 and the subsequent text.

Table 1. Summary of Rotaviruses occurrence data (adapted from da Silva et al., 2016, unless otherwise noted)

Source	Prevalence		Concentration (GC ^a /L unless otherwise noted)				Method
	Studies (samples)	% positive	Studies (samples)	Average	Studies (samples)	Range	
Raw Sewage	17 (1267)	8.4–100	NR ^c	NR	4 (153)	2.2×10 ² –2.9×10 ⁸	PCR ^d
	1 (46) ^b	72					PCR
Wastewater sludge	2 (24)	33–45	NR	NR	NR	NR	PCR
	1 (72) ^e	50	NR	NR	NR	NR	PCR
Treated wastewater	1 (36)	14–47	NR	NR	1(36)	6.5–16.6 PFU ^g /L	Culture
	2 (40)	ND ^f –100	1(16)	5.9×10 ⁵			PCR
Surface water	17 (1083)	14–100	NR	NR	1(55)	<3.0–54 FFU ^h /L	Culture
					6 (692)	ND–2.4×10 ⁷	PCR
					1(12)	0.19–8.3 PDU ⁱ /L	PCR
Groundwater	2 (122)	37–40	NR	NR	NR	NR	PCR
	1 (134) ^j	32					
	1 (11) ^k	27					
	1 (711) ^l	2.8					
Drinking-water	1 (730) ^m	16	NR	NR	NR	NR	PCR
	1 (370) ^e	0.54	NR	NR	NR	NR	PCR

^a genome copies

^b Santiso-Bellón et al., 2020

^c not reported

^d polymerase chain reaction

^e Kittigul & Pombubpa, 2021

^f not detected

^g plaque-forming units

^h focus-forming units

ⁱ infectious PCR detectable unit

^j Gamazo et al., 2018

^k Lutterodt et al., 2018

^l Pang et al., 2021

^m Kluge et al., 2014

6.1 Faecal pollution sources

Worldwide, rotavirus RNA was detected in 8.4% to 100% of raw sewage samples with concentrations of up to 2.9×10^8 GC/L (da Silva et al., 2016). In one study, 72% of sewage samples collected over a one-year period in Valencia, Spain, contained rotavirus RNA (Santiso-Bellón et al., 2020). In wastewater sludge, rotavirus RNA was detected in up to 50% of samples; concentrations were not reported (da Silva et al., 2016; Kittigul & Pombubpa, 2021). In a recent review, the prevalence of rotavirus in different sludge types, as determined by molecular methods, ranged from 25% to 83% (Gholipour et al., 2022). Rotaviruses are also frequently detected in treated wastewater. In one study, rotavirus RNA was detected in all samples from a municipal wastewater treatment plant following multiple treatment steps and disinfection, with a mean level of 5.9×10^5 GC/L (Qiu et al., 2015). Rotavirus were detected by molecular methods in 100% of treated wastewater effluent for Tianjin City, China (Shi et al., 2021).

Animal faeces are likely to be an important source of rotavirus strains in the environment as rotaviruses are a leading cause of diarrhoea in a long list of species, including pigs, calves, horses, rabbits, dogs, and turkeys among others (da Silva et al., 2016).

6.2 Surface water

Rotaviruses have been detected in surface waters world-wide. Rotavirus RNA was present in 14% to 100% of samples tested, in concentrations ranging from non-detectable to 2.4×10^7 GC/L (da Silva et al., 2016; Miura et al., 2019; Shi et al., 2021). One infectivity study reported a maximum of 54 focus-forming units/L in creeks (Mehnert & Stewien, 1993). The presence and concentrations of rotaviruses may depend on the season. In areas with temperate climates, they peak in the winter and spring months, but in tropical areas, this seasonality is not as marked (da Silva et al., 2016). It is estimated that a total of 2.0×10^{18} rotavirus particles are emitted into surface waters globally each year, of which 87% are produced by the urban human population (Kiulia et al., 2015).

Espinosa et al. (2008) demonstrated a 4 log₁₀ reduction in rotavirus concentrations measured in river water at 20 °C, pH 7.9, after 180 days. Another study reported a 2 log₁₀ reduction in rotavirus concentrations after 10 days in river water at 20 °C (da Silva et al., 2016).

6.3 Groundwater

In 2016, da Silva et al. summarized the existing data on the presence of rotavirus in groundwater. In one study in Slovenia, rotavirus RNA was detected in over one third of groundwater sources used for drinking-water (Steyer et al., 2011). In rural Bangladesh, rotavirus RNA was detected in ground water from 40% of tube wells (Ferguson et al., 2012). Since then rotavirus RNA has been detected in 32% of samples from an aquifer in Uruguay (Gamazo et al., 2018), 27% of hand-dug wells in Ghana (Lutterodt et al., 2018) and 2.8% of private wells in rural Alberta, Canada (Pang et al., 2021).

6.4 Drinking-water

In a study of treated tap water collected in public schools in Brazil, 16% of samples were positive for rotaviruses by molecular methods (Kluge et al., 2014). In Thailand, 0.54% of tap water samples were positive for rotavirus VP7 gene (Kittigul & Pombubpa, 2021). Rashid et al. (2021) detected genotype A rotavirus in 23% of tap water samples in Karachi, Pakistan. In

a study in Tianjin City, China, 92% of tap water samples were positive for rotavirus by molecular methods (Shi et al., 2021). In tap water held at 4 °C, there was no significant drop in rotavirus plaque titres even after 64 days; at 20 °C, there was a 2 log₁₀ reduction over the same period. Based on these and the surface water data, it can be concluded that rotaviruses persist for long periods of time in water (da Silva et al., 2016).²

6.5 Soil and crops

Both soil and crops can be contaminated with enteric viruses through irrigation with contaminated water or the application of organic waste. Rotaviruses have been detected with molecular methods on vegetables and in shellfish obtained directly from different aquatic environments such as marine farms and mangroves (da Silva et al., 2016).

6.6 Surfaces

Rotaviruses are found in moderate concentrations on surfaces and fomites. One study found that 14% (73/504) of surface samples from an adult intensive care unit in Brazil were contaminated with viral loads ranging from 3.4 GC/L to 2.9×10^6 GC/L (Ganime et al., 2012). Rotaviruses can persist for long periods on surfaces and be transferred from surfaces to hands (da Silva et al., 2016). When desiccated in a faecal suspension on a china surface at 4 °C, a 2 log₁₀ reduction in rotavirus infectivity, based on culture methods, was seen after about 15 days; rotaviruses then persisted up to 60 days with no further reduction. At 20 °C, a 2 log₁₀ reduction of rotaviruses was seen in less than 3 days. No further reductions in culturability were observed, which then remained stable for up to 60 days (Abad et al., 1994).

7 Significance of sanitation

Safe sanitation systems are important barriers against the spread of rotaviruses (da Silva et al., 2016). Outbreaks have been associated with inadequate sanitation leading to faecal contamination of drinking-water supplies. However, given the importance of person-to-person and fomite transmission (Estes & Greenberg, 2013), additional control measures including adequate hygiene, are needed to effectively prevent transmission. The importance of these additional measures is supported by evidence of worldwide prevalence of rotaviruses including in high-income settings where access to safely managed sanitation services is high (Levy et al., 2009; Patel et al., 2013). As early as 1985, it was noted that sites in Bangladesh and Canada had the same incidence of rotavirus infection despite dramatically different environmental and sanitation conditions (Esrey et al., 1985). Rotaviruses are highly infective and large concentrations of virions are dispersed into the local environment in faeces and vomit, suggesting the illness of one individual may easily be transmitted to others (Julian, 2016).

² Based on the criteria established by WHO in the Guidelines for drinking-water quality as follows: Detection period for infective stage in water at 20 °C: Short, up to 1 week; Moderate, 1 week to 1 month; Long, over 1 month.

8 Significance in drinking-water

Rotaviruses are major waterborne pathogens. There is strong evidence linking transmission of rotaviruses with unsafe drinking-water. Two waterborne outbreaks, involving a total of 13 311 cases, occurred in China between 1982 and 1983 where a novel rotavirus was identified as the etiologic agent (Tao et al., 1984). An outbreak in Colorado, United States of America, in 1981 affected 41 people and was linked to the breakdown in multiple water treatment processes that left no effective barrier between waste from toilets and the drinking-water supply (Hopkins et al., 1984). Rotavirus caused an outbreak of acute diarrhoeal disease involving 9907 cases in Turkey in 2005. At the time of the outbreak, the water supply had very low to undetectable levels of chlorine. The outbreak was halted when effective chlorination of the drinking-water supply was implemented, suggesting that the drinking-water was a source of transmission (Koroglu et al., 2011). In the Philippines in 2016, a rotavirus outbreak involving 2936 suspected cases was linked to water refilling stations (Rebato et al., 2019). A water sample from each of the five refilling stations where 138 case patients collected water were positive for rotavirus RNA. In a study in Tianjin City, China, rotavirus detected by molecular methods in two urban rivers were positively correlated with clinical prevalence (assessed by the presence of the viruses in stool samples) (Shi et al., 2021).

9 Prevention and management

Systematic approaches that identify and control faecal contamination along the whole sanitation service and drinking-water supply chains, are needed to manage microbial risks, including from rotaviruses.

The effectiveness of sanitation and drinking-water treatment systems depends on the selection and operation of technologies and the characteristics of the target organisms. Treatment processes based on size exclusion (e.g. filtration through media or soil in septic tank drainage areas) or settling in primary wastewater treatment processes will be limited for removing viruses, including rotaviruses. Rotaviruses are moderately resistant to oxidizing chemical disinfectants such as chlorine but have low resistance to disinfection by UV light.

9.1 Sanitation systems

Sanitation safety plans (WHO, 2022) should include control measures to safely collect, treat and discharge or reuse wastewater and faecal sludges. Removal or inactivation of rotaviruses by wastewater treatment processes is variable although the data are limited (de Silva et al., 2016). Based on available evidence and molecular methods, the reduction of rotaviruses by treatment processes is likely to be lower than the reduction of faecal indicator bacteria such as *E. coli*.

There are no data available on reduction removal of rotaviruses by on-site systems. In centralized sewage treatment plants secondary treatment achieved 0.8 to 2.5 log₁₀ reduction removal of rotaviruses (Kiulia et al., 2015). Wastewater treatment plants incorporating activated sludge and either membrane filtration or coagulation, precipitation and sand filtration achieved mean reductions of 2 log₁₀ to 2.8 log₁₀ (Li et al., 2011). In a study at a sewage treatment plant (Ibrahim et al., 2020), 100% of raw sewage samples, activated sludge- treated samples and UV

light-treated samples contained rotavirus A as measured by molecular methods. However, activated sludge and UV light treatments inactivated 85% and 73% of the rotavirus present, respectively. In laboratory trials submerged membrane bioreactors achieved a 5 log₁₀ reduction (Hmaied et al., 2015). In a full-scale membrane bioreactor trial, Miura et al. (2018) reported a 2 log₁₀ reduction in geometric mean concentrations of rotaviruses.

9.2 Drinking-water supplies

Within a water safety plan (WHO, 2012; 2023) control measures to manage potential risks from rotaviruses should include protection of source waters (surface or groundwater) from contamination by human faecal wastes, application of adequate treatment processes and protection of drinking-water quality during distribution and storage.

Removing rotaviruses by conventional drinking-water treatment processes (coagulation, flocculation, sedimentation and filtration) will likely be similar to that of other enteric viruses (Teunis et al., 2009). Moreover, conventional treatment can remove 1.5 log₁₀ of noroviruses (Shin & Sobsey, 2015), which are also non-enveloped RNA viruses of similar size as rotaviruses. Rotaviruses are moderately resistant to free chlorine³. For example, the Ct (product of disinfectant concentration and contact time) value for free chlorine for a 2 log₁₀ inactivation at 20 °C and a pH 7.2 for rotavirus strain Wa was 1.1 to 1.4 mg.min/L (Xue et al., 2013). Despite moderate resistance, chlorine can be effective provided adequate doses and contact times are applied. Rotaviruses are sensitive to UV light. A 20 mJ/cm² dose will achieve a 2 log₁₀ inactivation of rotavirus (Hijnen et al., 2006).

Due to the smaller size of rotaviruses relative to *E. coli*, resulting in lower removal by treatment processes such as filtration, and their greater environmental persistence, the absence of *E. coli* (or alternatively, thermotolerant coliforms) is not a suitable indicator of the absence of human rotaviruses in drinking-water. However, the presence of *E. coli* is a suitable indicator of the possible presence of human rotaviruses.

10 ICD-11 classification

XN6N7	Rotavirus
XN6TN	Rotavirus A
XN55H	Rotavirus B
XN0F5	Rotavirus C
XN29P	Rotavirus D
XN71N	Rotavirus E
1A22	Gastroenteritis due to Rotavirus

³ Within pathogen species and groups, there are likely to be variations in resistance, which could be further impacted by characteristics of the water supply and operating conditions. Resistance is based on the criteria established by WHO in the Guidelines for drinking-water quality as follows: 99% inactivation at 20 °C where, generally, Low represents a Ct₉₉ of < 1 min.mg/L, Moderate 1–30 min.mg/L and High > 30 min.mg/L (where C = the concentration of free chlorine in mg/L and t = contact time in minutes) under the following conditions: the infective stage is freely suspended in water treated at conventional doses and contact times, and the pH is between 7 and 8.

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Annex. Approach to content development and declarations of interest

The background documents were developed drawing from the following resources (i) microbial fact sheets from the fourth edition of the WHO Guidelines for Drinking-Water Quality (ii) pathogen-specific chapters in the Global Water Pathogen Project (GWPP) book “*Sanitation and Disease in the 21st Century*”, an update of the seminal book by Feachem et al. 1983 “*Sanitation and Disease: Health Aspects of Excreta and Wastewater Management*” and (iii) a rapid review of the literature at the start of the development of the background documents and before finalization in December 2023, focusing on identifying additional recent high quality literature reviews on a given pathogen. The documents were prepared by members of the WHO drinking-water quality microbiology group as well as external scientists with expertise in the relevant pathogens. The documents went through several iterations based on inputs from members of the WHO drinking-water quality microbiology working group and WHO technical staff. In addition, each document was reviewed by at least two external peer reviewers including authors of the GWPP book chapters. The extensive review process was designed to ensure technical accuracy and to identify any data gaps. The background documents were discussed at three expert meetings to resolve any issues and ensure the content reflected a consensus of expert opinions. Upon satisfactory resolution of any further data gaps or issues, the background documents went through technical editing before finalization.

Before the key expert meetings mentioned above, and for this work more broadly, all microbiology working group members submitted declarations of interest to WHO. These were to disclose any potential conflicts of interest that might affect, or might reasonably be perceived to affect, their objectivity and independence in relation to the subject matter of the guidance. WHO reviewed each of the declarations and concluded that none could give rise to a potential or reasonably perceived conflict of interest related to the subjects discussed at the meetings or covered by the guidance.